EDGE INTELLIGENCE
CASE STUDY II
VIDEO ANALYTICS

25MML0038

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1. Introduction

Edge Video Analytics (EVA) refers to performing video analysis tasks such as object detection, tracking, activity recognition, and anomaly detection directly on edge devices (near cameras), instead of transmitting raw video to centralized cloud servers.

With the exponential growth of video data from surveillance systems, IoT devices, and smart city applications, cloud-based processing faces challenges such as high latency, bandwidth consumption, and privacy risks. Edge computing overcomes these limitations by enabling real-time processing close to the data source, ensuring faster response and efficient resource utilization.

2. Edge Video Analytics Pipeline

A typical EVA system consists of the following components:

- Camera/Sensor Layer: Captures raw video frames
- Preprocessing at Edge: Frame sampling, resizing, ROI extraction
- Inference Engine: Deep learning models (YOLO, CNNs) deployed on edge devices
- Post-processing: Tracking, filtering, and event detection
- Selective Offloading: Only important metadata is sent to cloud

3. Case Study Analysis of Recent Research Papers (2024–2026)

3.1 SAMEdge: Edge-Cloud Video Analytics Architecture (2024)

Use Case: Smart surveillance and large-scale video analytics

- Problem:
  Large AI models like SAM require high computational resources, making them difficult to deploy directly on edge devices.
- Solution:
  SAMEdge introduces an edge-cloud collaborative architecture, where:
  - Edge handles lightweight tasks (detection, filtering)
  - Cloud performs heavy computations
- Key Contributions:
  - Efficient workload partitioning
  - Visual prompt transformation
  - Reduced latency with high accuracy
- Result:
  Enables deployment of advanced AI models in real-time edge environments

3.2 In-Vehicle Edge System for Real-Time Dashcam Video Analysis (2024)

Use Case: Autonomous driving and road safety

- Problem:
  Vehicles require real-time video analysis for hazard detection, but centralized systems introduce delay.
- Solution:
  A Distributed Edge Video Analytics (DEVA) system inside vehicles:
  - Uses multiple edge devices
  - Dynamically distributes workload
- Key Contributions:
  - Adaptive frame rate control
  - Real-time hazard detection
  - Distributed processing
- Result:
  Achieves 22–30 FPS with ~200 ms latency, suitable for real-time driving decisions

3.3 Edge AI: A Comprehensive Survey (2024)

Use Case: Smart cities, surveillance, and healthcare

- Problem:
  Traditional AI systems cannot handle real-time video analytics efficiently due to centralized processing.
- Solution:
  Edge AI enables:
  - Distributed processing
  - Parallel execution across edge devices
- Key Contributions:
  - Framework for deploying AI on edge
  - Scalability and low-latency processing
- Result:
  Provides a strong foundation for real-time intelligent systems

3.4 Cloud-Edge Collaborative Video Analytics Framework (2024)

Use Case: Traffic monitoring and smart transportation

- Problem:
  Balancing accuracy, latency, and bandwidth usage is challenging in video analytics systems.
- Solution:
  A cloud-edge collaborative model where:
  - Edge performs initial processing
  - Cloud handles complex analytics
- Key Contributions:
  - Dynamic resolution selection
  - Bandwidth optimization algorithm
- Result:
  Improves system efficiency and real-time responsiveness

3.5 Video Analytics in Cloud-Edge-Terminal Systems (2025)

Use Case: Smart cities and large-scale surveillance

- Problem:
  Single-layer systems (only cloud or only edge) are not scalable.
- Solution:
  A three-layer architecture (CETC):
  - Terminal (cameras)
  - Edge (local processing)
  - Cloud (global analysis)
- Key Contributions:
  - Hybrid architecture design
  - Adaptive scheduling
- Result:
  Improves scalability, performance, and privacy

3.6 Task-Oriented Communication for EVA (2024)

Use Case: Bandwidth-constrained environments

- Problem:
  Sending full video streams consumes high bandwidth.
- Solution:
  Transmit only task-relevant features instead of full frames
- Key Contributions:
  - Feature-based communication
  - Reduced redundancy
- Result:
  Significantly reduces bandwidth usage while maintaining performance

4. Key Insights from the Case Study

From the above research papers, the following insights are observed:

- Edge-cloud collaboration is essential for balancing computation and latency
- Lightweight AI models and optimization techniques are crucial for edge deployment
- Bandwidth-efficient communication methods improve scalability

- Privacy-preserving processing at edge is becoming a key requireme**nt**

5. Challenges

- Limited computation and memory in edge devices
- Energy constraints
- Security and privacy concerns
- Managing distributed systems

6. Real-World Use Case: Smart Traffic Monitoring System

Edge Video Analytics is widely used in smart traffic management systems to monitor and control road traffic in real time.

- Scenario:
  Traffic cameras installed at road intersections continuously capture video data.
- Problem:
  In traditional systems, sending full video streams to the cloud leads to:
  - High latency
  - Network congestion
  - Delayed traffic control decisions
- Solution:
  Edge devices placed near cameras process video locally and detect:
  - Vehicle count
  - Traffic density
  - Signal violations
    Only important information (alerts and statistics) is sent to the cloud.
- Technologies Used:
  - YOLO-based object detection models
  - Edge devices (e.g., NVIDIA Jetson, IoT gateways)
  - Real-time video analytics algorithms
- Result:
  - Faster traffic signal adjustments

        - Reduced traffic congestion
        - Real-time accident detection
        - Lower bandwidth usage
- Impact:
  Improves traffic efficiency, enhances road safety, and supports smart city infrastructure.

7. Conclusion

Edge Video Analytics is transforming modern intelligent systems by enabling real-time processing, reduced latency, and improved privacy.

The case study of recent research (2024–2026) demonstrates that:

- Advanced architectures like edge-cloud collaboration improve performance
- Distributed systems enable scalable analytics
- Optimization techniques make EVA practical for real-world applications

With ongoing advancements in Edge AI, deep learning, and communication technologies, EVA will play a crucial role in smart cities, autonomous systems, and industrial automation.

8. References

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